

STAT 36612: Randomness and High Dimensional Optimization

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Broadly speaking, we will consider problems of the form

$$\min_{x \in \mathcal{X}} f(x)$$

for (potentially) randomized f and $\mathcal{X}$ sufficiently large such that brute force is not feasible.

1 Examples

1. Consider the problem of finding the top eigenvalue of a random matrix G in the Gaussian orthogonal ensemble:

$$\mathbb{E} \max_{x \in \mathbb{R}^N, \|x\|_2=1} \langle x, Gx \rangle$$

with $G_{ij} = G_{ji} \sim N(0, 1/N)$, and $G_{ii} = 0$. Wigner's semicircle law shows that this is ≈ 2 for large N . Without the maximum, this problem is simple:

$$\mathbb{E}[\langle x, Gx \rangle] = \langle x, \mathbb{E}[G]x \rangle = 0$$

and

$$\begin{aligned} \text{Var}(\langle x, Gx \rangle) &= \mathbb{E} \left[\left(2 \sum_{i < j} x_i G_{ij} x_j \right)^2 \right] \\ &= 4 \mathbb{E} \left[\sum_{i < j} x_i^2 G_{ij}^2 x_j^2 \right] \\ &= \frac{4}{N} \sum_{i < j} x_i^2 x_j^2 \\ &= \frac{2}{N} \left(\left(\sum_{i=1}^n x_i^2 \right)^2 - \sum_{i=1}^n x_i^4 \right) \\ &= \frac{2}{N} (1 - \|x\|_4^4) \approx \frac{2}{N} \end{aligned}$$

so $\langle x, Gx \rangle \sim N(0, 2/N)$.

2. Alternatively, consider

$$\mathbb{E} \max_{x \in \{\pm 1\}^N} \langle x, Gx \rangle.$$

While harder, we can demonstrate that it is bounded above by $2N$ by reducing to the previous case:

$$\frac{1}{N} \mathbb{E} \max_{x \in \{\pm 1\}^N} \langle x, Gx \rangle = \mathbb{E} \max_{x \in \{\pm 1\}^N} \left\langle \frac{x}{\sqrt{N}}, G \frac{x}{\sqrt{N}} \right\rangle \leq \mathbb{E} \max_{x \in \mathbb{R}^N, \|x\|_2=1} \langle x, Gx \rangle \approx 2.$$

3. (A Random Convex Program) Take a (non-symmetric) random matrix $X \in \mathbb{R}^{n \times p}$ with $X_{ij} \sim N(0, 1)$ and

$$Y = X\beta + \xi$$

for $\xi \sim N(0, 1)$. Then we might try to ask, e.g.,

$$\min_{\hat{\beta} \in \mathbb{R}^p} \|Y - X\hat{\beta}\|_2^2 + \lambda \|\hat{\beta}\|_1$$

in hopes of recovering a sufficiently sparse β vector. Being able to compute properties of the achieved minimum can translate into computing properties of $\hat{\beta}$ (relative to the “true” underlying β).

4. (A Nonconvex/Combinatorial Problem) Let $A \in \mathbb{R}^{N \times N}$ with $A_{ij} \sim \text{Ber}(1/2)$, $A_{ii} = 1$. Then we might ask about

$$\mathbb{E} \max\{|S| \mid S \subset [N], A_{SS} = \mathbf{1}\}$$

which we call the largest clique problem (think of A as a random adjacency matrix for a graph). Further, we might care about limitations on “compute”: what can we do in polynomial time?

5. (An Application) One sends radio signals as waves; but there is usually some corruption between the sent and received messages. So how does one send messages over a noisy channel? The core math problem is as follows: given a message $M \in \{\pm 1\}^n$, is there a way to encode it into $\{\pm 1\}^N$ such that decoding is robust to corruption? For modern solutions, look for low density parity checks: write

$$Y = (M, Z) + \text{noise}$$

where Z is some additional information, and take Z_i to be the parity of the sum of a bunch of (well chosen) entries of M . Then decoding is a problem that looks like

$$\hat{M} = \operatorname{argmin}_{M \in \{\pm 1\}^n} \|Y - \text{enc}(M)\|_0.$$